

Project Proposal – Naïve Rag with image knowledge base

TechSupport RAG — Multi-Document & Image Troubleshooting Assistant

1. Problem Statement

Technical troubleshooting information is often scattered across multiple sources such as **user manuals, installation guides, troubleshooting PDFs, FAQs, and error screenshots/images**.

When a user faces a problem, they have to manually search through several documents to identify:

- What the error means
- Possible causes
- Troubleshooting steps
- Required settings
- Warning/error messages
- Relevant diagrams or screenshots

This makes troubleshooting time-consuming, especially for users who are not technically experienced.

Proposed Solution

Develop a **Multi-Document + Image Knowledge Base using Naive RAG** that allows users to ask troubleshooting questions in natural language.

The system will use multiple **PDF documents and PNG/JPG images** as its knowledge base. Relevant information will be retrieved from the knowledge base and provided to an LLM to generate a concise, context-based troubleshooting response.

2. Example Scenario

Imagine a **Wi-Fi Router Troubleshooting Assistant**.

Knowledge Base:

knowledge_base/

|

|— router_user_manual.pdf

|— installation_guide.pdf

└─ troubleshooting_guide.pdf

└─ faq.pdf

|

└─ router_lights.png

└─ error_message.png

└─ network_diagram.png

User asks:

"My router's internet light is blinking red. What should I do?"

The system retrieves relevant information from:

troubleshooting_guide.pdf

+

router_lights.png

+

user_manual.pdf

The LLM then generates:

The red Internet indicator generally indicates a connectivity problem. Check that the WAN cable is properly connected, restart the router, and verify the connection status. If the indicator remains red, follow the ISP connection troubleshooting steps in the guide.

3. Project Objective

The objective is to build a **Retrieval-Augmented Generation (RAG) based troubleshooting assistant** that can:

1. Accept multiple PDF documents as a knowledge base.
2. Accept PNG/JPG images containing useful troubleshooting information.
3. Extract text from documents.
4. Extract meaningful information from images using a vision-capable LLM.
5. Convert the extracted content into embeddings.
6. Store the knowledge in a vector database.
7. Retrieve relevant information based on a user's question.

8. Generate an answer using the retrieved context.
9. Inform the user when the required information is not available.

Example Knowledge Base

A practical student project could use **Laptop Troubleshooting**.

Documents

File	Content
laptop_manual.pdf	Laptop specifications and controls
installation_guide.pdf	Setup and installation
troubleshooting.pdf	Common problems and solutions
maintenance_guide.pdf	Maintenance instructions
faq.pdf	Frequently asked questions

Images

Image	Information
error_screen.png	Error message screenshot
indicator_lights.png	LED/status indicators
port_diagram.png	Laptop port diagram
boot_error.png	Boot error screen

4. Sample User Queries

The system should handle questions such as:

"My laptop is not charging. What should I check?"

"What does this red indicator mean?"

"I am getting this error message. How can I fix it?"

"Which port should I use to connect the external monitor?"

"My laptop is overheating. What troubleshooting steps are recommended?"

"The laptop does not boot after pressing the power button. What should I do?"

5. Technology Stack

Programming

Python

Document Processing

- pypdf
- RecursiveCharacterTextSplitter

Image Understanding

- Vision-capable LLM

Embeddings

- OpenAI text-embedding-3-small

Vector Database

- ChromaDB

LLM

- GPT-4o-mini

Framework

- LangChain

Optional UI

- Streamlit

6. Suggested Project Structure

TechSupport-RAG/

|

├─ app.py

├─ requirements.txt

├─ .env

|

├─ knowledge_base/

```
| |— manuals/
| | |— user_manual.pdf
| | |— installation_guide.pdf
| | |— troubleshooting.pdf
| |
| |— images/
| | |— error_screen.png
| | |— indicator_lights.png
| | |— port_diagram.png
|
|— chroma_db/
```

7. Expected Output

Instead of simply returning retrieved chunks, the system should provide a structured troubleshooting response:

Problem Identified:

Internet connection issue

Possible Cause:

WAN connection may be disconnected.

Recommended Steps:

1. Check the WAN cable.
2. Restart the router.
3. Check the Internet indicator.
4. Verify ISP connectivity.

Reference:

Troubleshooting Guide

